

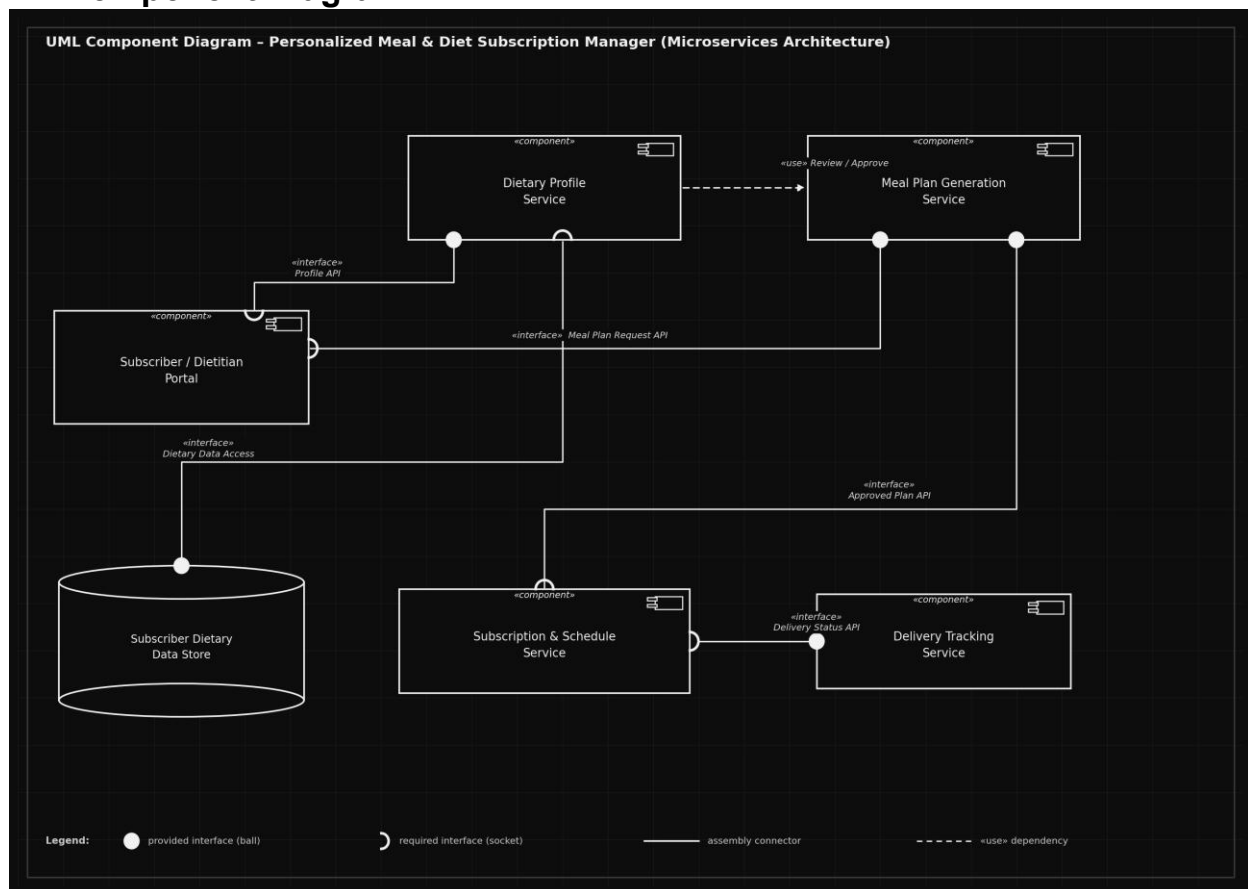
SOFTWARE ENGINEERING

Lab 3: Component Modelling & Architectural Pattern Selection Problem Statement #19 – Personalized Meal & Diet Subscription

Manager

Name: Rushikesh Patil
SRN: PES1UG24AM394
SEC: G

1. Component Diagram



2. Architectural Justification

Architecture Selection

We chose Microservices Architecture for the Personalized Meal & Diet Subscription Manager System.

Architectural Choice

The system is divided into focused services for dietary profile management, meal-plan generation, subscription scheduling, delivery tracking, and subscriber dietary data.

Each service communicates through well-defined interfaces instead of keeping all functions in one large application.

Reason 1 – Separation of Core Functions

Meal planning, dietary information, subscription changes and delivery tracking have different responsibilities. Keeping them as separate services reduces coupling and allows changes in one function without changing the complete subscription workflow.

Reason 2 – Schedule and Delivery Handling

The Subscription & Schedule Service can handle pause, skip and reschedule requests independently, while the Delivery Tracking Service records daily fulfilment status. This supports the 12-hour modification requirement before dispatch.

Security Advantage

NFR-002 requires subscriber dietary and allergy information to be protected from unauthorized access. Keeping this information behind a controlled Subscriber Dietary Data Store limits direct access and allows access rules to be applied at the service boundary.

Performance Benefit

Meal-plan generation and subscription-schedule processing can be scaled independently. During periods with many requests, only the busy service needs additional resources, helping valid schedule modifications to be processed within the required 12-hour cutoff.